

Hybrid force/impedance control method for ultrasound robot based on three-dimensional modeling

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Abstract—The ultrasound robot not only needs to be able to maintain stable contact forces in the direction of the scan check normal in unknown or complex environments, but also needs to maintain safe human-robot interaction forces in other directions, and at the same time needs to satisfied the constraints of avoiding joint limits. In response to the above control requirements, this paper proposes a hybrid force/impedance control method for ultrasound robots based on three-dimensional modeling, which performs indirect force impedance control in the scan tangential direction of the ultrasound robot, and direct force control in the scan normal direction maintains stable contact forces between the ultrasound probe and human tissue, and introducing environmental three-dimensional point cloud information for scan normal position compensation. This improves the force control performance of the ultrasound robot in unknown environments. The tangential impedance control reduces the tangential contact force generated by the scanning motion of the robot and prevents the physical injury to the patient. The weighted minimum parametric number is also introduced into the hybrid force/impedance control to optimize the motion of the null space of the ultrasound robot and avoid the ultrasound robot from entering the joint limit in the contact interaction control. Finally the effectiveness and reliability of the proposed method is verified by ultrasound robot experiments.

Keywords—ultrasound robot, impedance control, joint limit avoidance, force control, three-dimensional modeling

I. INTRODUCTION

Ultrasonography is widely used in the diagnosis of internal organs and superficial organs due to its non-invasive, non-radiation, simple operation and low cost characteristics^[1]. However, during the ultrasound examination, the doctor needs to hold the probe to scan the area to be examined on the human body and maintain the same posture for a long time, which may cause muscle fatigue or even damage, thereby reducing the accuracy of ultrasound diagnosis. The use of ultrasound robot can significantly improve the accuracy and reproducibility of ultrasound examinations^[2].

The ultrasound robot needs to be in contact with the patient for a long time while accurately replicating the scanning action of the doctor. The compliance control of the robot can protect the personal safety to the greatest extent. At the same time, the contact conditions of the ultrasound probe have a significant impact on the quality of the obtained

ultrasound images^[3]. Therefore, maintaining a safe and stable contact interaction between the ultrasound probe and human tissue is a key technology for robotic ultrasound scanning. Existing studies usually use algorithms such as impedance control^[4-5], admittance control^[6], and hybrid force-image^[7] to deal with this problem. Jay Carriere^[8] used an admittance control algorithm on a semi-automatic robotic ultrasound system to maintain stable contact force under external disturbance conditions. Fang^[9] used an admittance force control algorithm on a collaboratively controlled six-degree-of-freedom robotic ultrasound system to make the robotic arm conform to the operator's movement, and assist in applying force during the scanning process to reduce the force that the doctor needs to use. Napoli^[10] introduced a force-velocity hybrid control algorithm in robotic ultrasound scanning by decomposing the task space in six Cartesian directions and applying the force control algorithm in the direction normal to the contact surface. Abbas M^[11] combined force and position hybrid control to help the operator keep the probe centered within its range of motion.

With the expansion of ultrasound scanning application scensess, the requirements for the flexibility of robot are also increasing, and more and more researchers tend to choose redundant robotic arms to conduct related research on ultrasound robot^[12]. Due to the existence of the null space, the redundant manipulator can improve its kinematic and dynamic performance by optimizing the self-motion in the null space of the robot. The introduction of redundancy also increases the difficulty of controlling the robot.

Therefore, the research on the redundancy characteristics of ultrasound robot is also a research difficulty and hot spot in this field. Ficuciello^[13] established an optimization strategy for the redundant solution of the manipulator in the Cartesian impedance controller in the experimental study of the human-machine cooperative operation with motion redundancy, and increased the secondary position and velocity control of the singular position in the null space of the task space effector. However, the complicated calculation amount also increases the difficulty of real-time control of the robot, and this method cannot guarantee the effective control of the contact force. Aiming at the joint vibration phenomenon of the WAM manipulator used in auxiliary nursing due to the singularity of the fifth and seventh joints, David Parent^[14] designed the speed performance function of the fifth and seventh joints as the null-space projection

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moment, and realized the null-space under a specific joint configuration. Optimized for singularity avoidance. Although a lot of results have been achieved in the study of safe human-machine contact interaction control and redundancy characteristics of ultrasound robot, there are still many factors that limit their further promotion and application. The main difficulty is that ultrasound robot need to be able to maintain stable contact forces in the direction of the scan normal to get accurate ultrasound images in an unknown scanning environment, but also need to maintain safe interaction forces in other directions, and at the same time need to optimize the motion of the robot's null space to avoid dangerous positions such as joint limits during ultrasound scanning by ultrasound robot.

In this paper, we propose a hybrid force/impedance control method for a 7-DOF ultrasound robot based on three-dimensional modeling. By performing indirect force impedance control in the scanning tangential direction of the ultrasound robot, while maintaining a stable contact force between the ultrasound probe and the human tissue by direct force control in the scanning normal direction, while using the environmental three-dimensional point cloud model information as compensation for the environmental information in the scanning normal direction, the force control performance in the dynamic environment is improved, and the tangential impedance control reduces the tangential contact force generated by the scanning motion of the robot. And the weighted minimum parametric number is introduced into the hybrid force/impedance control to optimize the motion of the null space of the ultrasound robot and finally the effectiveness and reliability of the proposed method is verified by the ultrasound robot experiment.

II. ANALYSIS OF CONTACT FORCE BETWEEN ULTRASOUND ROBOT AND HUMAN BODY

When the ultrasound robot is working, the end of the robot is equipped with a medical ultrasound probe to scan the surface of human organs and visceral tissues and skin. To complete a standard ultrasound scan and image acquisition, it is necessary to ensure the correct placement of the ultrasound probe and stable tissue contact force.



Figure.1. Ultrasound robot abdominal ultrasound scan scene

The autonomous ultrasound robot jointly developed by our team and Hefei Hebin Intelligent Robot Company adopts a seven-degree-of-freedom high-performance torque control robot arm, and its working scenes of human abdominal ultrasound scanning is shown in Figure 1, Where the X and Y axes of the Cartesian task space at the end of the robot form the tangential motion plane of the ultrasound scanning.

The contact between the end probe tool and human tissue surface during ultrasound scanning will generate a large frictional force in the opposite direction of the tangential movement of ultrasound scanning, which will easily cause human skin pain and even accidental abrasions or contusions and other personal damages. Therefore, the ultrasound robot needs to be softly controlled in the scanning tangential direction to reduce friction. And in the robot scanning normal Z-axis need to carry out direct force control, to provide stable contact force between the ultrasound probe and human tissue surface, to reduce the ultrasound image black area, to ensure the quality of ultrasound image acquisition. For the flexibility of the ultrasound robot requirements, while the ultrasound robot in the process of scanning movement needs to ensure the safety of the configuration, to avoid the robot into the joint limit or odd position caused by joint movement too fast to produce a greater psychological pressure on the patient.

Force control between the probe and human skin is a matter of patient safety and the quality of ultrasound images. Existing research methods use impedance control in the direction of the contact force of the ultrasound robot for indirect force control on the human tissue uncertainty complex surface and deformation characteristics, it is difficult to achieve a precise and stable force tracking effect, can not guarantee the standard acquisition of ultrasound images. The use of hybrid force/position control algorithms reduces the patient experience in the scanning cut-up, while common control methods do not provide motion control of the ultrasound robot in null spaces.

We propose a hybrid force/impedance control method for a 7-DOF ultrasound robot based on three-dimensional modeling. When the ultrasound robot performs the scanning task, the method maintains an impedance relationship in the tangential direction of the scanning by indirect force control, and maintains a stable expected force between the scanning normal probe and human tissue by direct force control, while introducing environmental three-dimensional point cloud information as position compensation in the scanning normal direction, and using a hybrid friction model for the dynamics of the ultrasound robot to improve the normal force tracking accuracy. In addition, this method can avoid the ultrasound robot from entering the joint limit during the scanning task.

III. HYBRID FORCE/IMPEDANCE CONTROL METHOD BASED ON THREE-DIMENSIONAL MODELING

The control block diagram of the weighted least parametric 7-DOF ultrasound robot avoidance joint limit mixing and force/impedance control method is shown in Figure.2. The robot uses a high-precision joint torque controller, and the control torque τ_c output by the algorithm to the robot joint torque controller is as follows:

$$\tau_c = \tau_F^Z + \tau_Y^Z + \tau_{WLN}^{XY} + C(\dot{q}, q)\dot{q} + G(q) + F(\dot{q}) \quad (1)$$

where τ_F^Z is the joint torque input corresponding to the scan normal force controller, τ_Y^Z is the joint torque corresponding to the environmental information compensation, τ_{WLN}^{XY} is the joint torque input corresponding to the ultrasound robot scan tangential impedance controller

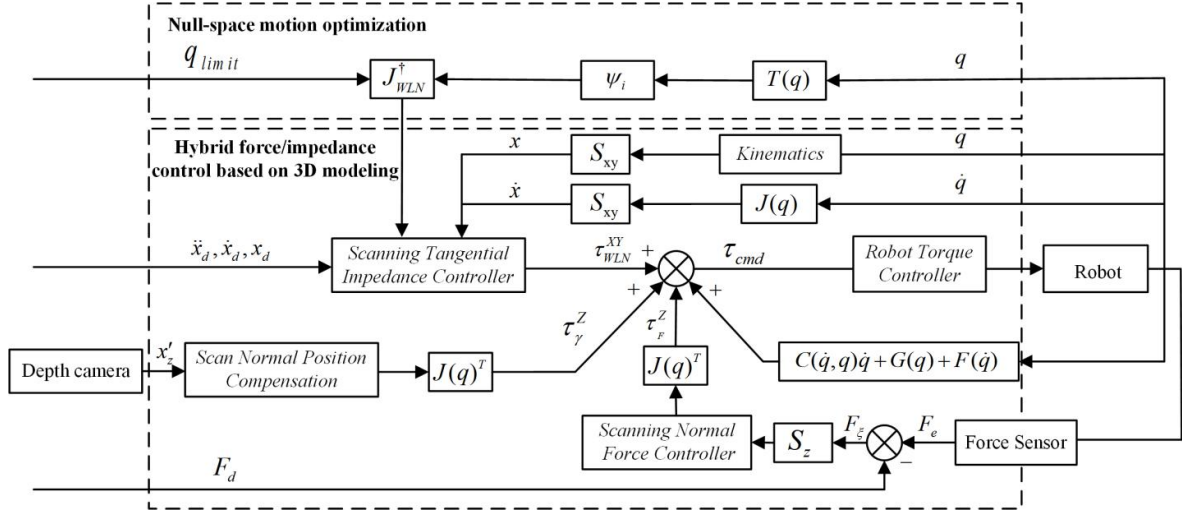


Figure 2. Hybrid force/impedance control based on three-dimensional modeling block diagram

$C(\dot{q}, q)\dot{q}$ is the robot Coriolis force and centrifugal force

vector, $G(a)$ is the gravity vector and $F(\dot{a})$ is the joint friction force.

A. Normal position compensation based on three-dimensional modeling

The goal of three-dimensional modeling is to capture the 3D shape appearance of objects and scenes in high quality to enable simulated three-dimensional perception and interaction in digital space. Depth camera-based three-dimensional modeling tends to be more robust to environmental disturbances. In this paper, we use Intel's three-dimensional vision sensor realsense D435i installed in the ultrasound robot end effector, the depth camera plays the role of perceiving the outside in ultrasound robot our system.

The RGB-D depth camera installed at the end of the ultrasound robot acquires three-dimensional point cloud data containing color information, and performs matching calibration and noise reduction rendering of the point cloud data set to obtain an optimized three-dimensional point cloud model. The three-dimensional point cloud model of the surface platform identified by the depth camera at the end of the ultrasound robot is shown in Figure 3.

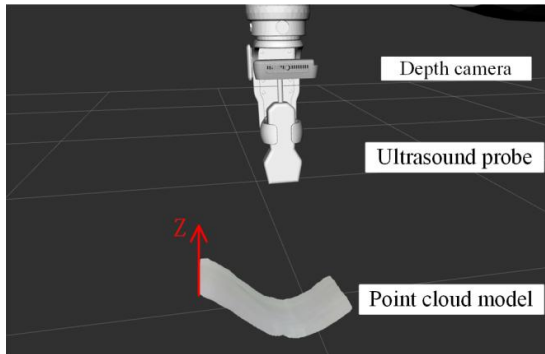


Figure 3. scanning the normal contact surface point cloud model

In order to improve the tracking accuracy of the normal scanning force of the ultrasound robot in the unknown and complex environment, this paper designs the normal environmental compensation torque to compensate the uncertainty caused by the unknown environment and model

error by introducing the depth information of the collected three-dimensional point cloud model of the environment in the scanning normal direction as the environmental information compensation of the ultrasound scanning task.

Its corresponding compensating torque τ_F^z as

$$\tau_F^z = J^T(q)(K_z(x'_z - x_z) + D_z(\dot{x}'_z - \dot{x}_z)) \quad (2)$$

where $x'_z, \dot{x}'_z$ are the input quantities of the scan method to the environmental information, and K_z, D_z are the corresponding proportional and differential gain matrices, respectively.

B. Hybrid force/impedance control

The ultrasound robot is designed with a PI controller in the scan normal direction for direct force control to maintain stable contact of the probe, corresponding to the torque control input τ_F^z as

$$\tau_F^z = J^T(q)S_z(F_d + k_{fp}F_\xi + k_{fi} \int F_\xi(t)dt) \quad (3)$$

$$F_\xi = F_d - F_e \quad (4)$$

where F_ξ is the robot scan normal contact force deviation, F_d and F_e are the scan normal desired contact force and actual contact force respectively^[15]. k_{fp} and k_{fi} denote the proportional and integral gains, respectively. S_z is the scan normal force vector selection matrix with diagonal factor vectors of $[001000]^T$.

For the dynamics of the ultrasound robot in this paper, the following hybrid friction model is designed

$$F(\dot{q}) = f_c \text{sgn}(\dot{q}) + f_v \dot{q} + (f_s - f_c)e^{-(\dot{q}/v_s)^\delta} \quad (5)$$

Where f_c is the coefficient of Coulomb friction, f_v is the coefficient of viscous friction, v_s is the stribek velocity, and δ is the empirical constant.

The dynamic properties of the second-order system consisting of mass-spring-damping embodied at the end of the robot in the tangential direction of the ultrasound scan are as follows:

$$M_d \ddot{\tilde{x}} + D_d \dot{\tilde{x}} + K_d \tilde{x} = F_e \quad (6)$$

where $\tilde{x}, \dot{\tilde{x}}, \ddot{\tilde{x}}$ are the deviations of robot position, velocity and acceleration from the desired values, respectively, and M_d, K_d, D_d are the desired mass, stiffness and damping matrices, respectively. Assuming that the planned scan trajectory is $\ddot{x}_d, \dot{x}_d, x_d$, the joint torque input to maintain the tangential impedance of the scan can be defined as

$$\begin{aligned} \tau^{XY} = & M(q)J^{-1}M_d^{-1}(M(q)\ddot{x}_d + K\tilde{x} + D\dot{\tilde{x}} - M_d\dot{J}\dot{q}) \\ & + (J^T - M(q)J^{-1}M_d^{-1})F_e \end{aligned} \quad (7)$$

Considering the limitation of the torque sensor, the above control rate can be designed $M_d = J(q)^{-T}M(q)J(q)^{-1}$ simplified so that τ^{XY} and F_e are independent and the robot can still reflect the impedance characteristics, but the coupling exists with the mass-related term.

The traditional impedance method lacks motion control for redundant robot in null space, this paper introduces weighted minimum parametric for robot null space motion optimization, defined as follows based on the weighted minimum parametric robot Jacobi pseudo-inverse J_{WLN}^+ substitution in equation (7) for J^{-1} .

$$J_{WLN}^+ = \begin{bmatrix} \psi_1 & & \\ & \ddots & \\ & & \psi_n \end{bmatrix}^{-1} J^T (J \begin{bmatrix} \psi_1 & & \\ & \ddots & \\ & & \psi_n \end{bmatrix} J^T)^{-1} \quad (8)$$

where ψ_i positive definite weighted factor is defined as

$$\psi_i = \begin{cases} 1 + |\partial T(q) / \partial q_i| & \Delta |\partial T(q) / \partial q_i| \geq 0 \\ 1 & \Delta |\partial T(q) / \partial q_i| < 0 \end{cases} \quad (9)$$

$$T(q) = \sum_{i=1}^n \frac{1}{4} \frac{(q_{i\max} - q_{i\min})^2}{(q_{i\max} - q_i)(q_i - q_{i\min})} \quad (10)$$

Where $T(q)$ is robot avoidance joint limit optimization criterion function, $q_{i\max}, q_{i\min}$ are the upper and lower limits of the joint respectively. The definition of ψ_i in equation (9) shows that its value will tend to infinity at the joint limit and to 1 at the center position. when the robot joint is close to its limit position, the corresponding null space weight factor will increase, thus limiting the motion of the corresponding joint. when the robot has a tendency to

move away from the current joint limit position, it will not limit its joint motion.

The complete form of the corresponding torque τ_{WLN}^{XY} of the proposed scan tangential impedance controller in this paper is as follows.

$$\begin{aligned} \tau_{WLN}^{XY} = & M(q) \begin{bmatrix} \psi_1 & & \\ & \ddots & \\ & & \psi_n \end{bmatrix}^{-1} J^T (J \begin{bmatrix} \psi_1 & & \\ & \ddots & \\ & & \psi_n \end{bmatrix} J^T)^{-1} \\ & M_d^{-1} (M_d \ddot{x}_d + K S_{xy} \tilde{x} + D S_{xy} \dot{\tilde{x}} - M_d \dot{J} \dot{q}) \end{aligned} \quad (11)$$

where S_{xy} is the scanning tangential position velocity vector selection matrix with a diagonal factor vector of $[110000]^T$.

IV. EXPERIMENTAL VALIDATION

A. Null-space motion optimization effectiveness experiment

To verify the effectiveness of joint avoidance limit null space optimization in the proposed control method, a 7-DOF ultrasound robot is used as the experimental platform to experimentally control the robot end probe motion trajectory as a linear motion along the tangential Y-axis of the scan check. The acceleration in the Y-axis direction is $0.04m/s^2$, the maximum speed is $0.04m/s$, and the running time is 5 seconds. The initial joint angle of the ultrasound robot is $[7.4, 12.6, -6.8, 108.8, 3.4, 165, 43.5]$ deg.

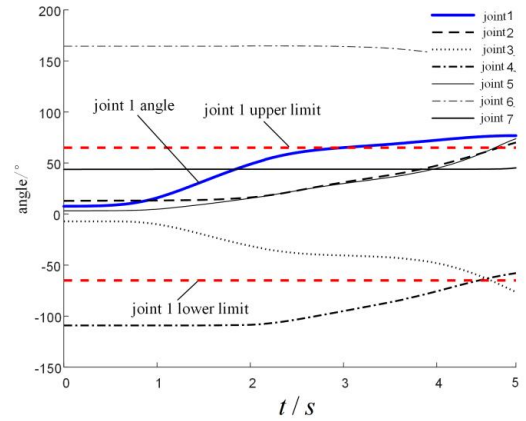


Figure.4(a) no joint avoidance limit optimization

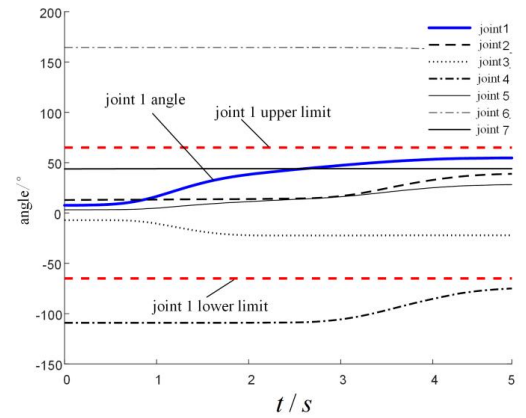


Figure.4(b) joint avoidance limit optimization

Figure.3. comparison of angles before optimization of joint avoidance limits

Figure 4(a) shows the joint angles before joint limit avoidance optimization; then the robot joint limits are set to upper limit as $[65, 80, 160, 160, 160, 170, 160]$ deg and lower limit as $[-65, -80, -160, 0, -160, -170, -160]$ deg. Figure 4(b) shows the joint angles after joint limit avoidance optimization in null space, as shown in Figure, the joint angles remain within the limits after optimization, while all other joints of the robot remain within the set limits.

B. Ultrasound robot hybrid force control experiment

To verify the effectiveness and reliability of the force control algorithm of the hybrid force/impedance control method for ultrasound robot based on three-dimensional modeling proposed in this paper in unknown or complex environments that may be encountered when contacting with human tissues, force tracking control experiments were conducted in three different force contact environments namely, inclined, curved and medical mimetic models, using the hybrid force control method without normal three-dimensional environment information compensation. Finally, force tracking control experiments were conducted in curved environments using the hybrid force/impedance control method for ultrasound robot based on three-dimensional modeling proposed in this paper and compared with the original control method.

During the experiment, the end of the ultrasound robot is controlled to hold the ultrasound probe down and maintain the desired contact force F_d (shown as the red line in Figure 5) for a distance along the tangential Y-axis of the ultrasound scan. In addition, the tangential X-axis contact force F_x and the tangential Y-axis contact force F_y are measured, and since the displacement of the end probe along the X-axis is very small in the experimental task, only the Y-axis contact force F_y^* under tangential position control is considered for comparison.

The experimental results are shown in Figure 5, where Figure 5(a), Figure 5(b) and Figure 5(c) show the force tracking control experimental scenes and the observation results of the robot end force of the ultrasound robot on the inclined experimental platform, the curved experimental platform and the medical profiling model without considering the normal environmental information compensation, respectively. The average tracking error of the scan normal contact force on the inclined experimental platform is 0.63 N, and the maximum contact force error is 1.75 N. The average tracking error of the force on the curved experimental platform is 0.67 N, and the maximum contact force error is 2.67 N. The average tracking error of the force on the medical mimetic model is 0.3 N, and the maximum contact force error is 0.72 N. In addition, Figure 5(c) shows on the upper left the ultrasound robot. The ultrasound image of the left lobe of the liver acquired on the medical mimetic model.

Figure 6(a) shows the experimental scene of force tracking control on the curved experimental platform with the proposed hybrid force/impedance control method considering the compensation of normal three-dimensional environment information, Figure 6(b) shows the simulation scene of three-dimensional point cloud model, and Figure 6(c) shows the observation results of the end force of the ultrasound robot. The average force tracking error on the curved platform is 0.49N and the maximum tracking error is 1.95N, which can achieve better force tracking performance than that without environmental information compensation. Meanwhile, the tangential contact forces under tangential impedance control of the ultrasound robot are all significantly smaller than those under position control.

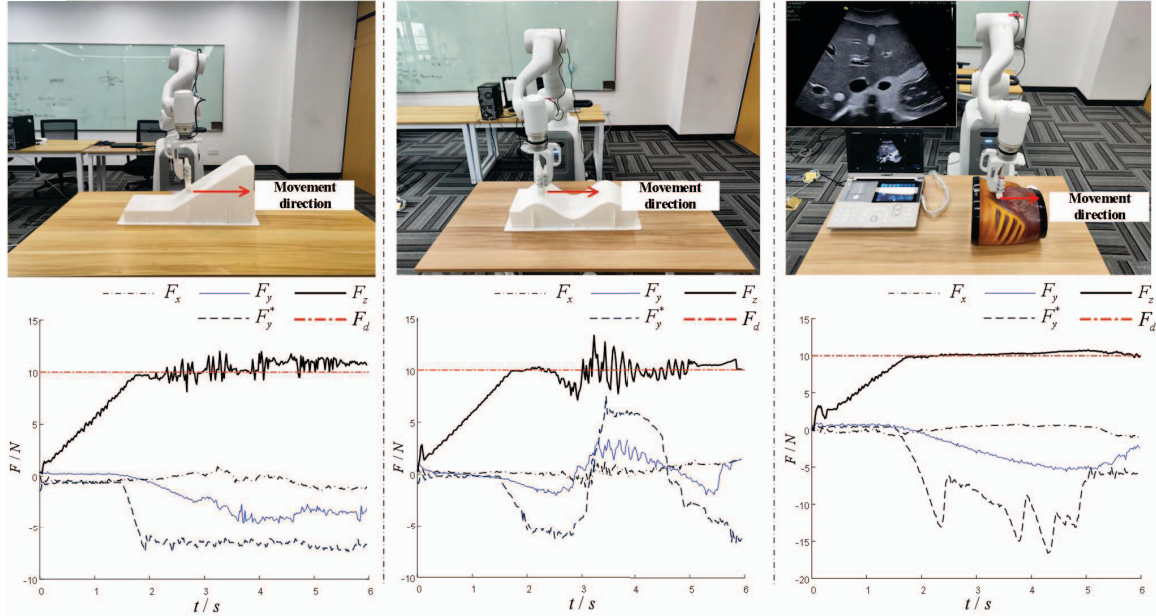


Figure.5(a).

Figure.5(b).

Figure.5(c).

Figure.5. ultrasound robot hybrid force control experiment

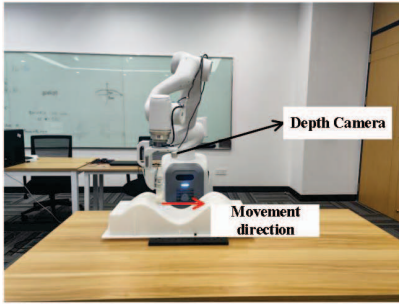


Figure.6(a).

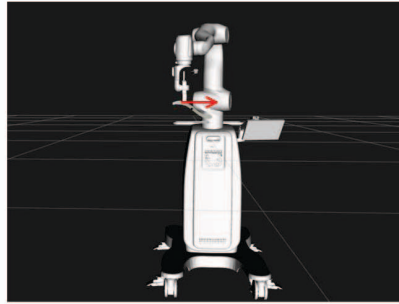


Figure.6(b).

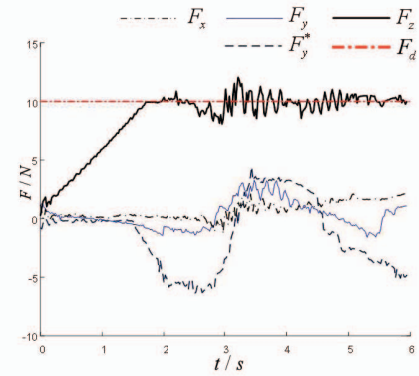


Figure.6(c).

Figure.6. ultrasound robot hybrid force control based on three-dimensional modeling experiment

V. CONCLUSION

In this paper, we propose a hybrid force/impedance control method for a 7-DOF ultrasound robot based on three-dimensional modeling, which performs impedance control in the scanning tangential direction of the ultrasound robot and introduces normal three-dimensional environment point cloud information compensation in the scanning normal direction to be able to maintain stable contact forces. At the same time, the weighted minimum parametric number is introduced into the hybrid force/impedance control to optimize the motion of the null space of the ultrasound robot, and the effectiveness of the joint avoidance limit optimization is demonstrated through experiments. Finally, the experimental results show that the average tracking error of the scanning normal contact force on the inclined surface is 0.63N and the average tracking error of the curved force is 0.67N without the introduction of three-dimensional environmental information compensation, while the average tracking error of the proposed method on the curved force is 0.49N after the introduction of three-dimensional environmental information compensation. At the same time, the scanning tangential contact force is significantly smaller than the contact force under position control, which meets the demand of tangential impedance control in human-computer interaction.

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